Computer Communication and Networks

Session Layer

Session 17

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Network Layer

Learning Objectives: At the end of the lesson, you are expected to be able to:

- Dialogue Management
- Synchronization
- Activity Management
- Exception Handling



Session Layer

Session Layer:

- The session layer is the 5th layer of OSI model.
  - ✓ It responds to service requests from the presentation layer
  - ✓ It provides service requests to the transport layer.

- The session layer provides the mechanism for opening, closing and managing a session between end-user application processes
  - ✓ That is, a semi-permanent dialogue.

- Communication sessions consist of requests and responses that occur between applications.



Session Layer

Session Layer:

- Session layers are commonly used in application environments that make use of remote procedure calls (RPCs).
  - ✓ In case of a connection loss this protocol may try to recover the connection.
  - ✓ If a connection is not used for a long period, the session layer protocol may close it and re-open it.

- It provides for either full duplex or half-duplex operation and provides synchronization points in the stream of exchanged messages.



Session Layer: Services

Session Layer: Session

- Think of a session as a conversation that must be kept separate from others to prevent intermingling of the data.
- Three steps to set up a Session:
  1. Setting up a Session
  2. Maintaining a Session
  3. Tear Down a Session



Session Layer: Services

Session Layer: Setting Up a Session

- To setup a session:

1. Check user credentials
2. Assign numbers to session to identify them
3. Negotiate services needed for session
4. Negotiate who begins sending data



Session Layer: Services

Session Layer: Maintaining a Session

- To maintain a session:

1. Transfer the data
2. Re-establish a disconnected session
3. Acknowledging receipt of data



Session Layer: Services

Session Layer: Tear Down a Session

- To tear a session:

1. Due to mutual agreement → once the transfer is done
2. Due to other party disconnected.



Session Layer: Services

Session Layer: Services to the Transport Layer

- Session Layer provides following services:

1. Dialogue Management
2. Synchronization
3. Activity Management
4. Exception Handling



Session Layer: Services

Service: Dialogue Management

- Dialogue Management allow two systems to **communicate.**
  - ✓ Deciding whose turn it is to talk.

- Some applications operate in half-duplex mode, whereby the two sides alternate between sending and receiving messages, and never send data simultaneously.

- In the ISO protocols, dialog management is implemented through the use of a data token.
  - ✓ The token is sent back and forth, and a user may transmit only when it possesses the token.



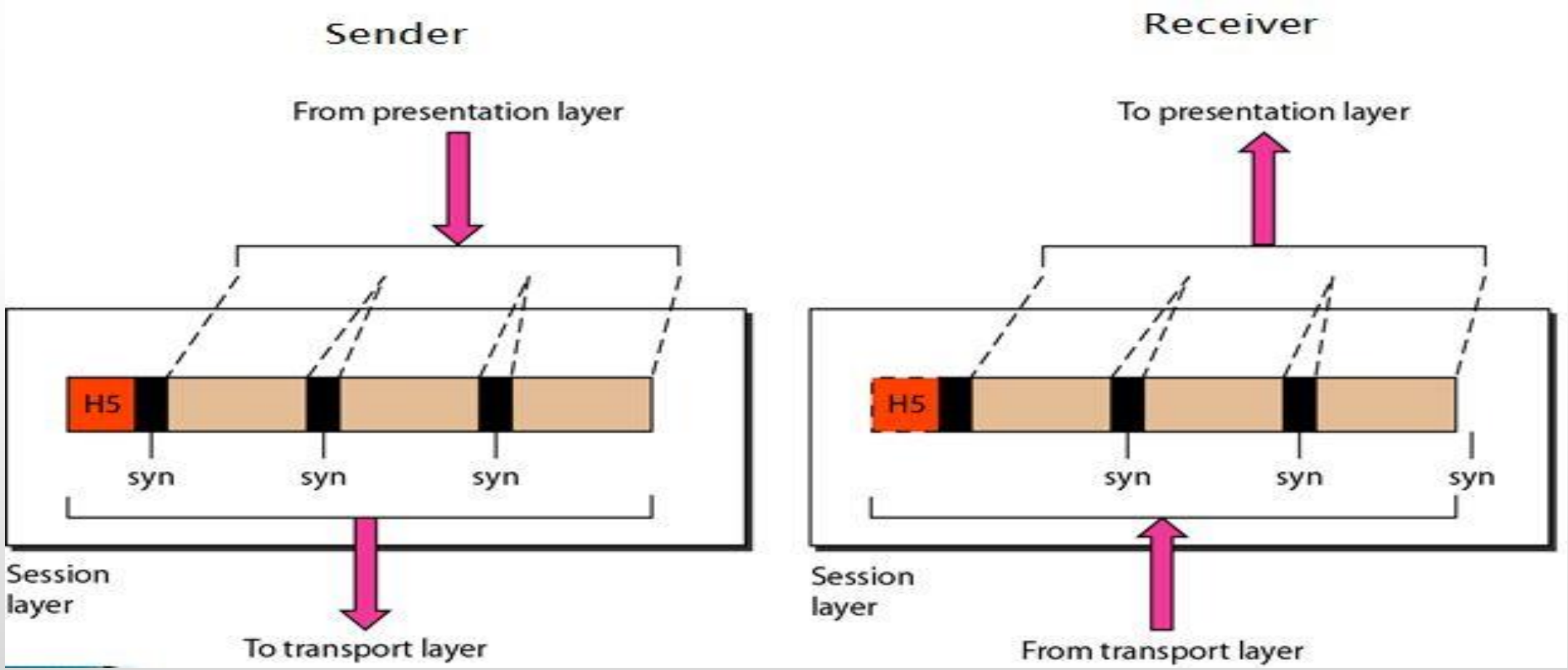
Session Layer: Services

Service: Synchronization

- The transport layer handles only communication errors, synchronization deals with upper layer errors.
  - For example, in a file transfer, the transport layer might deliver data correctly, but the application layer might be unable to write the file because the file system is full.
  - Users can split the data stream into pages, inserting synchronization points between each page.

Session Layer: Services

❖ Synchronization





Session Layer: Services

Service: Synchronization cont.

- When an error occurs, the receiver can resynchronize the state of the session to a previous synchronization point.
  - ✓ Move the two session entities into a known state.
- Synchronization is achieved through the use of **sequence numbers**.
- There are two types of synchronization:
  1. Minor Synchronization point
  2. Major synchronization point



Session Layer: Services

Service: Activity Management

- Each activity is independent of activities that come before and after it, and an activity can be processed on its own.
- Uses of Activity:
  1. Used to delimit files of a multi-file transfer
  2. Used for quarantining:
     - ✓ collecting all the messages of a multi-message exchange together before processing them.



Session Layer: Services

Service: Activity Management Cont.

- The receiving application would begin processing messages only after all the messages had arrived.
- This provides a way of helping insure that all or none of a set of operations are performed.
  - ✓ For example, a bank transaction may consist of locking a record, updating a value, and then unlocking the record.
  - ✓ If an application processed the first operation, but never received the remaining operations (due to client or network failures), the record would remain locked forever. Quarantining addresses this problem



Session Layer: Services

Service: Exception handling

- A General purpose mechanism for reporting errors.
  - ✓ The service provider can generate an Exception-Report indication to notify the user about internal problems within the session layer or problems reported to it from the transport or lower layers.
  - ✓ These reports contain a field describing the nature of the exception.
  - ✓ It is up to the user to decide what action to take, if any.



References

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